1 Chapter 4

Exercise 4.1 In Example 4.1, if π is the equiprobable random policy, what is $q_\pi(11, down)$? What is $q_\pi(7, down)$?

For $q_\pi(11, down)$, we transition to S' = S^+ with reward -1 and a future expected reward of 0. Thus, q $q_\pi(11, down)$ = -1.

For $q_\pi(7, down)$, we transition to S' = 11 with reward -1 and a future expected reward of $-\gamma 14$. γ is just 1. Thus, $q_\pi(7, down)? = -15$.

Exercise 4.2 Suppose a new state 15 is added to the gridworld just below state 13, and its actions, left, up, right, and down, take the agent to states 12, 13, 14, and 15, respectively. Assume that the transitions from the original states are unchanged. What, then, is $v_\pi(15)$ for the equiprobable random policy? Now suppose the dynamics of state 13 are also changed, such that action down from state 13 takes the agent to the new state 15. What is $v_\pi(15)$ for the equiprobable random policy in this case?

$v_\pi(15)$ = -1 + $\frac{1}{4}(-22 - 20 - 14 + v_\pi(15))$ =>
$v_\pi(15) = \frac{4}{3}(-1 - 14) = -20$.

For part two of this question, we need to look at $v_\pi(13)$ and $v_\pi(15)$:
$v_\pi(13)$ = -1 + $\frac{1}{4}(-22 - 20 - 14 + v_\pi(15))$ =>
$v_\pi(13) - \frac{1}{4}v_\pi(15) = -15$

and from the first part of the question we have:
$\frac{3}{4}v_\pi(15) = -1 + \frac{1}{4}(-22 + v_\pi(13) - 14)$ =>
$\frac{3}{4}v_\pi(15) - \frac{1}{4}v_\pi(13) = -10$.

This is a system of linear equations, and doing the math out:

$\frac{4}{5}v_\pi(15) - 4 - \frac{1}{4}v_\pi(15) = -15$ =>
$v_\pi(15) = -20$.
Intuitively, this makes sense as the dynamics and returns are the same from each state $v_\pi(9)$ also yields -20; you can even simply retrace the arrows on the right hand side under "optimal policy" and the arrows in terms of direction and steps until termination are the same.

Exercise 4.3 What are the equations analogous to (4.3), (4.4), and (4.5) for the action-value function q_π and its successive approximation by a sequence of functions $q_0, q_1, q_2, ...$?

4.3) $q_\pi(s,a) = E[R_{t+1} + \Sigma_{a_{t+1}} \pi(a_{t+1}|s_{t+1}) \gamma q_\pi(s_{t+1}, a_{t+1})|s,a]$
4.4) $q_\pi(s,a) = \Sigma_{s',r'} p(s',r|a,s)[r + \Sigma_{a'} \pi(a'|s') \gamma q_\pi(s',a')]$
4.5) $q_{k+1}(s,a) = \Sigma_{s',r'} p(s',r|a,s)[r + \Sigma_{a'} \pi(a'|s') \gamma q_k(s',a')]$

Exercise 4.4 The policy iteration algorithm on page 80 has a subtle bug in that it may never terminate if the policy continually switches between two or more policies that are equally good. This is ok for pedagogy, but not for actual use. Modify the pseudocode so that convergence is guaranteed.

I think there are a few ways to fix this. One way would to see if the policy values themselves are within some some small threshold θ_{thresh} despite following different actions. If they are within the threshold, then just set policy-stable = true. The other way I am thinking is to look back two actions. That is, since we are seeing if action a is better than a-1, then we can see if action a-2 is equal to action a, and if yes, we know we are looping between these two actions and can set policy-stable = true.

Exercise 4.5 How would policy iteration be defined for action values? Give a complete algorithm for computing q_*, analogous to that on page 80 for computing v_*. Please pay special attention to this exercise, because the ideas involved will be used throughout the rest of the book.

1. Policy is arbitrarily initiated as before

2. Policy Evaluation

Loop:
  $\Delta \leftarrow 0$
  Loop for each s $\in$ S:
    $q \leftarrow Q(s,a)$
    $Q(s,a) \leftarrow \Sigma_{s',r'} p(s',r'|s,a)[r + \gamma \Sigma_{a'} \pi(a'|s') Q(s',a')]$
    $\delta \leftarrow max(\Delta|q - Q(s,a)|)$
until $\Delta < \theta$(a small positive number determining the accuracy of estimation)

3. Policy Improvement

policy-stable $\leftarrow$ *true*
For each $s \in S$:
  old-action $\leftarrow \pi_s$
  $\pi(s) \leftarrow argmax_a \Sigma_{s',r'} p(s',r'|s,\pi(s,a))[r + \gamma \Sigma_{a'} \pi(a'|s') Q(s',a')]$
  If old-action $\neq \pi(s,a)$, the policy-stable $\leftarrow$ *false*
If *policy-stable*, then stop and return $Q \approx Q_*$ and $\pi \approx \pi_*$; else go to 2.

Exercise 4.6 Suppose you are restricted to considering only policies that are ϵ-soft, meaning that the probability of selecting each action in each state, s, is at least $\frac{\epsilon}{|A(s)|}$. Describe qualitatively the changes that would be required in each of the steps 3, 2, and 1, in that order, of the policy iteration algorithm for v_* on page 80.

Every action now has a minimum chance of being selected given by $\frac{\epsilon}{|A(s)|}$. Policy is now stochastic. The policy can be arbitrarily initialized as before, but control statements must be implemented so as to guarantee the minimum probability for all actions and account for this on other actions so that $P_{total} = 1$ still. The new policy evaluation and improvement will be:

2. Policy Evaluation

Loop:
  $\Delta \leftarrow 0$
  Loop for each s $\in$ S:
    $v \leftarrow V(s)$
    $V(s) \leftarrow \Sigma_a \pi(a|s) \Sigma_{s',r'} p(s', r'|s, a)[r + \gamma V(s')]]$
    $\delta \leftarrow max(\Delta | V - V(s)|)$
until $\Delta < \theta$(a small positive number determining the accuracy of estimation)

3. Policy Iteration

policy-stable $\leftarrow$ *true*
For each $s \in S$:
  old-action $\leftarrow \pi_s$

$$\pi(s) = \begin{cases} 1 \text{ - } \epsilon + \frac{\epsilon}{|A(s)|}, \text{ if } a = argmax_a \text{V(s)} \\ \frac{\epsilon}{|A(s)|}, \text{ otherwise} \end{cases}$$

  If old-action $\neq \pi(s)$, the policy-stable $\leftarrow$ *false*
If *policy-stable*, then stop and return $V \approx V_*$ and $\pi \approx \pi_*$; else go to 2.

In words, when we find the argmax(a) for state s, we no longer can select a with probability 1. Instead, we must assign it the max probability under our ϵ-soft constraint. For example, if there are five states total, $\frac{\epsilon}{5}$ must be the minimum probability assigned to each state. Now once argmax(a) is found and we know p(a) is no longer 1, and since we still want to maximize p(a), we subtract: 1 - $\epsilon + \frac{\epsilon}{5} = 1 - \frac{4}{5}\epsilon$. $\frac{4}{5}\epsilon$ is the total probability of actions not including argmax(a) under our ϵ-soft constraint.

Exercise 4.8 Why does the optimal policy for the gambler's problem have such a curious form? In particular, for capital of 50 it bets it all on one flip, but for capital of 51 it does not. Why is this a good policy?.

$P_h = 0.4$, so if you bet 1.00 each round, you are expected to slowly lose. Hence, the 50.00 mark acts as a threshold: if you are at 50.00, you bet everything to minimize the amount of rounds played to win and thus maximize the probability of winning. Now if you are above 50.00 but below 75.00, you have the threshold of 50.00 to fall back on. You thus want to stake anything that you could reach to 75.00 with but keep you at or above 50.00, which is the reason for the small triangular ”tics”, which is the class of optimal policies referred to in the above paragraph (for example, at 65, you could arbitrarily stake 10 or 15). For the spike at 75, the logic is similar; we can win by betting 25.00, but if we lose we fall back to 50 (and if we are above 75 we can stake arbitrarily to win, but if we lose keep ourselves at or above 75). Likewise for 25: we bet 25.00 on the chance we jump to 50.00 and then bet 50.00 if we make it (and again, the tics are because if we are under 25 we can arbitrarily stake to get up to 25, and if we are above 25 but below 50, we can arbitrarily stake to get up to 50 but stay at or above 25 if we lose). To drive the point home again, we want to minimize the expected number of rounds played to win.

Exercise 4.10 What is the analog of the value iteration update (4.10) for action values, q k+1 (s, a)?

$$q_{k+1}(s,a) = \Sigma_{s',r'} p(s',r'|s,a)[r + \gamma max_{a'} q_k(s',a')]$$

Sutton and Barton Exercises Chapter 4

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2 Introduction